Behavior Poisoning in Cooperative MAPPO

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Key finding: Behavior poisoning was not reliably harmful; effects were seed-sensitive, mixed, and non-monotonic in p.

1. Motivation and Research Question

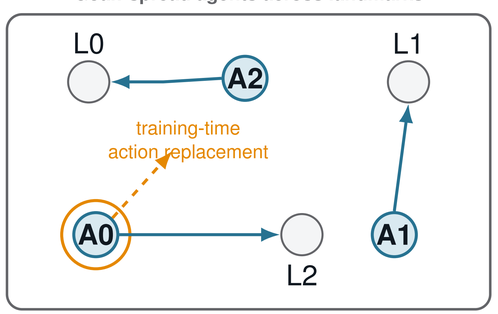
In cooperative MARL, training data is not independent: each agent's learning signal depends on trajectories produced jointly with teammates. A compromised teammate can alter the training distribution for the whole team without touching rewards, observations, gradients, or weights. This is subtle because the attack disappears at evaluation; any effect must be learned from poisoned training rollouts.

Research question

Can action-level behavior poisoning during MAPPO rollout collection degrade final clean-evaluation performance in `simple_spread`?

Task. In `simple_spread`, three agents cover three landmarks while avoiding unnecessary collisions. Good coordination spreads agents across distinct landmarks.

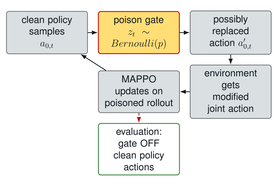
Goal: spread agents across landmarks



A0 training action may be replaced Eval: gate OFF

Core idea. Training-time action replacement → changed rollout distribution → seed-dependent clean-evaluation outcome.

2. Threat Model and Attack Mechanics



$$a'_{0,t} = \begin{cases} \text{poison}(a_{0,t}), & z_t = 1, \\ a_{0,t}, & z_t = 0. \end{cases}$$

Hook. At each rollout step, MAPPO samples $a_{0,t}$, the gate samples $z_t \sim Bernoulli(p)$, and agent 0's action is replaced only when $z_t = 1$. The environment receives the modified joint action; MAPPO updates on that trajectory. Evaluation sets $z_t = 0$, so positive D is learned, not test-time sabotage.

Attacker capability. Replace agent 0's training-time action only; no access to observations, rewards, parameters, gradients, update rule, or evaluation actions.

Observed outcome. The attacker wants $D > 0$, but D is not consistently positive in the sweep.

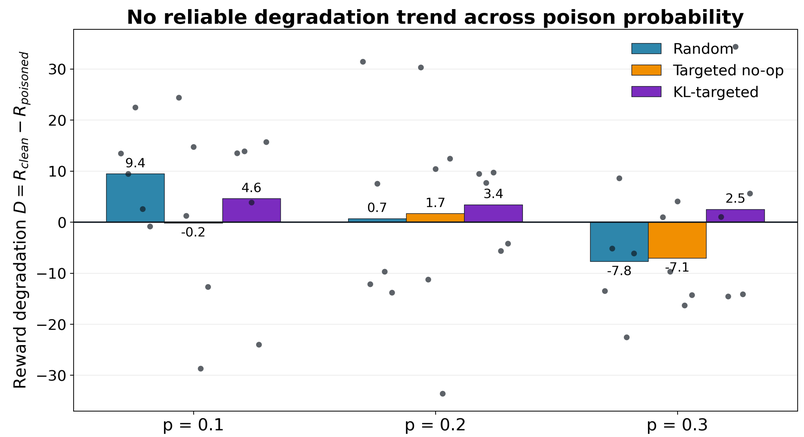
3. Attack Variants and Experimental Grid

Experimental grid. 5 clean baselines + 3 attacks $\times$ 3 poison probabilities $\times$ 5 seeds = 50 matched training runs. Each checkpoint uses 32 deterministic clean evaluation episodes. Matched seeds compare poisoned(seed s) against clean(seed s).

Attack	Replacement behavior	What it tests
Random	uniform legal action	Does unstructured teammate noise shift learning?
Targeted no-op	force action 0	Does a passive teammate induce bad coordination?
KL-targeted	prefer no-op under KL budget	Can a constrained attack create positive D?

4. Metric and How to Read Degradation

5. Main Result: Mixed Effects, Not Reliable Degradation



Caption. Positive D means the poisoned policy evaluated worse than the matched clean policy; negative D means it evaluated better. Random and no-op are not reliably harmful and become negative on average at $p = 0.3$.

p	Random	Targeted no-op	KL-targeted	Any $D > 0$	All $D > 0$
0.1	9.43	-0.21	4.58	5/5	3/5
0.2	0.65	1.66	3.39	4/5	2/5
0.3	-7.75	-7.06	2.47	3/5	1/5

Any $D > 0$ = at least one attack degrades that matched seed. All $D > 0$ = all three attacks degrade that matched seed.

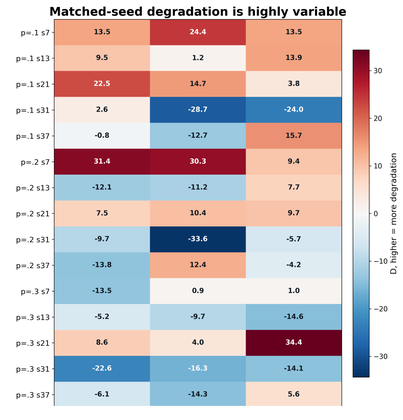
What the plot supports

- Some settings degrade: several attack/p averages have $D > 0$.
- Degradation is not reliable: random and no-op become beneficial on average at $p = 0.3$.
- Increasing p does not increase degradation.
- KL-targeted has positive mean D at all tested p, but not stable per-seed outcomes.

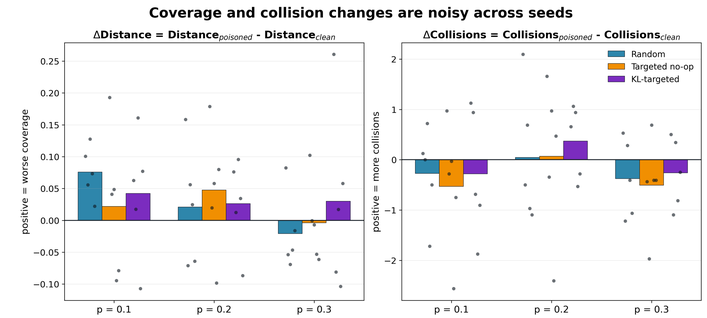
Answer and caution

Answer: sometimes, but not reliably. Positive examples exist, but many matched settings show no degradation or improvement. The data support mixed outcomes, not stable attack success.

6. Seed Sensitivity



7. Mechanism: Coverage and Collision Effects Are Noisy



Caption. Coverage and collision changes overlap across attacks and p values. These plots do not reveal a simple mechanism that consistently explains reward degradation.

- Reward includes both landmark coverage and collision penalties.
- Some poisoned policies improve one component while worsening another.
- Overlap is consistent with seed- and trajectory-dependent outcomes, but does not prove a mechanism.

Future defense ideas

If future work finds stronger behavior-poisoning attacks, likely defenses would monitor rollout behavior rather than only rewards or gradients.

- Action-distribution anomaly detection.
- Per-agent entropy or intervention monitoring.
- Robust rollout filtering and adversarial teammate training.

8. Hypothesis Check

Hypothesis	Outcome	Evidence
H1 Clean MAPPO learns coordination	Baseline established	clean runs provide matched baselines
H2 Poisoning reliably degrades clean evaluation	Not supported	many settings have $D \leq 0$; random/no-op are negative at $p = 0.3$
H3 No-op beats random	Not supported	no-op does not reliably produce higher D
H4 KL-constrained poisoning reliably degrades	Weak/mixed support	positive mean D at all tested p, but variable by seed
H5 More poison means more degradation	Not supported	average degradation decreases or reverses for random/no-op

Bottom line

Supported: non-monotonicity and seed sensitivity. Partially supported: some settings degrade. Not supported: reliable degradation, targeted dominance, or monotonic harm.

9. Takeaways, Limitations, and Next Steps

Takeaways

- Action-level poisoning is a plausible training-time threat model, but this sweep did not show reliable degradation.
- Some matched-seed settings degrade clean-evaluation reward; many show no degradation or improvement.
- The most robust finding is non-monotonic, seed-sensitive behavior rather than stable attack success.

Limitations: one environment; one compromised agent; three p values; one no-op target; five seeds; no confidence intervals. Conclusions are exploratory.

Next steps: add seeds and confidence intervals; search target actions; compare against benign noise; test whether KL averages persist.

Main lesson

The attack model is plausible, but observed degradation is intermittent rather than reliable.